From Nothing to Coulomb's Law

Electricity — lesson notes · discussion of 12 July 2026 · Dad's notes, corrected and extended

The chain of ideas

1. **Everything is made of particles.**
   Matter is made of atoms, and atoms are made of protons, neutrons and electrons.
2. **Electric charge is a property of a particle.**
   Just as a particle has mass, it can carry charge. Charge is measured in coulombs (C).
3. **Particles can be positive, negative, or neutral.**
   The proton is positive, the electron is negative, the neutron is neutral.
4. **Charge is quantised: it comes in steps of the elementary charge e.**
   Every observed charge is a whole-number multiple of e. The proton carries exactly +e and the electron exactly −e. (Fine print: quarks carry ±e/3 and ±2e/3, but they are never seen alone.)
5. **Charged particles interact: they attract or repel each other.**
   Neutral particles feel no electric force.
6. **Like charges repel; unlike charges attract.**
   + and + push apart, − and − push apart, + and − pull together.
7. **The net charge of a body counts its imbalance of particles: $Q = (N_+ - N_-) \cdot e$.**
   N_+ = number of positive charges (protons), N_- = number of negative ones (electrons). Most bodies have $N_+ \approx N_-$, so they are neutral.
8. **Charge is conserved.**
   Charge is never created or destroyed — it only moves from one body to another (e.g. rubbing a balloon moves electrons onto it).
9. **Coulomb's law tells us how strong the force is.**
   See the box below — the climax of the chain.

Coulomb's law

$$F = k \cdot \frac{|q_1| \cdot |q_2|}{r^2} \quad \text{with} \quad k = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \times 10^9 \ \text{N·m}^2/\text{C}^2$$

- **q_1, q_2** — the two charges (in coulombs); **r** — the distance between them (in metres). Valid for **point charges** (small compared with r).
- The force acts **along the line joining the charges**: like charges are pushed apart, unlike charges pulled together. Both charges feel the **same** force, in opposite directions (Newton's third law).
- **Inverse square:** double the distance → one quarter of the force.

Worked example. Two charges of 1 µC each, 10 cm apart: $F = 8.99\times10^9 \cdot (10^{-6})^2 / (0.1)^2 \approx$ **0.9 N** — about the weight of a small apple.

Scale check: two charges of 1 C at 1 m would push with 9×10^9 N (≈ the weight of 900 000 tonnes) — that is why real charges are microcoulombs.

Constants worth remembering

Quantity	Symbol	Value	Note

Elementary charge	e	1.602×10^{-19} C	exact: $1.602\ 176\ 634 \times 10^{-19}$ C
Electron charge	$-e$	-1.602×10^{-19} C	proton: $+e$
Coulomb constant	$k = 1/(4\pi\varepsilon_0)$	8.99×10^{9} N·m²/C²	remember as $\approx 9 \times 10^{9}$
Vacuum permittivity	ε_0	8.854×10^{-12} C²/(N·m²)	same thing as F/m
Charges in 1 coulomb	1 C / e	6.24×10^{18}	1 C is a *huge* charge
Electron mass	m_e	9.11×10^{-31} kg	proton is ≈1836× heavier

English fixes from the draft

In the draft	Correct
dipell / dispell	repel
magnitue	magnitude
charge particles	charged particles
equal signs of charge …	like charges …
opposite – attract	unlike (opposite) charges attract
Coulomb law: k*q1*q2/(r*r)	Coulomb's law: $F = k\cdot\lvert q_1\rvert\cdot\lvert q_2\rvert / r^2$

Where to read (from our archive)

Siyavula, Physical Sciences Gr 10 —
docs/textbook/siyavula-physical-sciences/Gr10_PhysicalSciences_Learner_Eng.pdf

ch 16 Electrostatics — §16.2 Two kinds of charge (PDF p.276), §16.3 Conservation of charge (PDF p.282), §16.4 Quantisation of charge (PDF p.284). Closest match to these notes.

CK-12, Physical Science Workbook —
docs/textbook/ck12-physical-science/CK-12-Physical-Science-MS-Workbook-with-Answers.pdf

lesson 23.1 Electric Charge (printed p.276 ≈ PDF p.283) — practice worksheets; answers in ch 26.23 (printed p.417).

NCERT, Class 9 Science —
docs/textbook/ncert-class9-physics/other-chapters/ch08-journey-inside-the-atom.pdf

§8.2–8.6 (PDF pp.2–12) — protons & electrons, atomic number: why atoms are neutral and where the charges sit.

Cambridge Lower Secondary, Stage 7 —
docs/textbook/cambridge/Cambridge_Lower_Secondary_Science_2ed_7_Learners_book.pdf

Unit 9, §9.1 Flow of electricity (≈ p.287) — moving charge = current: the next step after these notes.

The fully quantitative Coulomb's-law chapter is in **Siyavula Physical Sciences Grade 11** (free, openly licensed) — not in the archive yet; easy to add.